

Titanic Data Analysis – Insights & Conclusion

1. Project Objective

The objective of this project is to analyze the Titanic passenger dataset and understand the factors associated with passenger survival. The analysis focuses on variables such as gender, passenger class, age, family relationships, fare and embarkation point. Tableau was used to create visualizations and a dashboard to make the results easier to understand.

2. Dataset Overview

The dataset contains information about 891 Titanic passengers. It includes details such as passenger class, name, gender, age, number of siblings or spouses, number of parents or children, ticket fare, cabin and embarkation point.

The Survived column was used as the main outcome for the analysis, where 1 represents survived and 0 represents did not survive.

3. Overall Survival

Out of 891 passengers, 342 survived and 549 did not survive. This gives an overall observed survival rate of approximately 38.4%.

This means that fewer than half of the passengers in the dataset survived.

4. Survival by Gender

Gender showed a clear difference in survival.

There were 314 female passengers, of whom 233 survived, giving an observed survival rate of approximately 74.2%.

There were 577 male passengers, of whom 109 survived, giving an observed survival rate of approximately 18.9%.

Therefore, female passengers had a much higher observed survival rate than male passengers in this dataset.

5. Survival by Passenger Class

Passenger class also showed a clear difference in survival.

The observed survival rates were:

- 1st Class: 63.0%
- 2nd Class: 47.3%
- 3rd Class: 24.2%

First-class passengers had the highest observed survival rate, while third-class passengers had the lowest.

This shows a strong association between passenger class and survival in the dataset.

6. Survival by Age

There were 177 passengers with missing age information.

Among passengers with recorded ages:

- Under 18: 54.0% survived
- 18 and above: 38.1% survived

The under-18 group therefore had a higher observed survival rate than the 18-and-above group.

However, because some age information was missing, this result should be interpreted with some caution.

7. Survival by SibSp

SibSp represents the number of siblings or spouses travelling with a passenger.

The analysis showed:

- SibSp = 0: 34.5% survived
- SibSp > 0: 46.6% survived

Passengers who had at least one sibling or spouse aboard had a higher observed survival rate than passengers who were travelling without a sibling or spouse.

8. Survival by Parch

Parch represents the number of parents or children travelling with a passenger.

The results were:

- Parch = 0: 34.4% survived
- Parch > 0: 51.2% survived

Passengers travelling with at least one parent or child had a higher observed survival rate.

9. Survival by Fare

The median fare in the dataset was 14.45.

Passengers were divided into two groups:

- Lower Fare (≤ 14.45): 25.2% survived
- Higher Fare (> 14.45): 51.2% survived

The higher-fare group had a considerably higher observed survival rate.

However, fare is also related to passenger class, so this relationship should not be considered completely independent of passenger class.

10. Survival by Embarked Location

The dataset contains three main embarkation locations:

- C – Cherbourg: approximately 55.4% survived
- Q – Queenstown: approximately 39.0% survived
- S – Southampton: approximately 33.7% survived

Passengers who embarked at Cherbourg had the highest observed survival rate among the three locations.

There were also 2 passengers with missing embarkation information.

11. Missing Values

The missing-value check identified:

- Age: 177 missing values
- Cabin: 687 missing values
- Embarked: 2 missing values

The original dataset was kept unchanged rather than replacing missing values with assumptions. This helped preserve the original information in the dataset.

12. Tableau Dashboard

Tableau was used to present the main findings through visualizations and a dashboard.

The final dashboard includes:

- Overall Survival
- Survival by Gender
- Survival by Passenger Class
- Survival by Age Group
- Survival by Embarked

Additional worksheets were created for SibSp, Parch and Fare analysis.

The dashboard makes it easier to compare different passenger groups and identify patterns in survival.

13. Key Findings

The main findings from the analysis were:

1. The overall observed survival rate was approximately 38.4%.
2. Female passengers had a much higher observed survival rate than male passengers.
3. First-class passengers had a higher observed survival rate than second- and third-class passengers.
4. Passengers under 18 had a higher observed survival rate than passengers aged 18 and above among passengers with recorded ages.
5. Passengers travelling with siblings/spouses or parents/children generally had higher observed survival rates.
6. Passengers paying higher fares had a higher observed survival rate.
7. Survival rates also differed according to embarkation location.

14. Conclusion

The Titanic dataset shows that survival was not evenly distributed among passengers. Gender and passenger class showed particularly noticeable differences in survival rates. Age, family-related variables, fare and embarkation location also showed differences between passenger groups.

This project helped me understand the basic process of data analysis, including checking data, calculating survival rates, identifying missing values, creating groups and presenting findings through Tableau visualizations.

The results show associations within the dataset and do not prove that any single factor directly caused survival or death.

15. Tools and Skills Used

Tools:

- Tableau
- Spreadsheet analysis

Skills:

- Data cleaning and checking
- Missing-value analysis
- Basic data analysis
- Data grouping and comparison
- Data visualization
- Dashboard development
- Insight generation

Dataset: Titanic – Machine Learning from Disaster, Kaggle